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# Printability Metrics in Direct Ink Writing: Critical Review of the Literature and Novel Perspective Based on Dimensional Analysis

*Enrica Luzzi, Martina de Salzano de Luna, Domenico Caputo, Giovanni Filippone,* and Paolo Aprea*

**Direct Ink Writing (DIW) is a versatile additive manufacturing technique widely used for processing complex inks, particularly particle-loaded formulations. Despite its broad applicability, the field lacks standardized criteria for defining and assessing printability. Existing approaches, typically based on rheological thresholds such as yield stress and elastic modulus, often lead to system-specific, inconsistent results and rely heavily on subjective visual inspection. This work presents a systematic framework for analyzing DIW printability, grounded in the unpacking of the printing process into five subfunctions: extrudability, single filament accuracy, planar accuracy, buildability, and ability to produce suspended structures. Each subfunction is examined in detail through dimensional analysis to identify the parameters responsible for its success. This approach enables a deeper understanding of the interactions between material properties and process variables and highlights how certain parameters can have opposing effects in different subfunctions. To support an objective evaluation, a standardized printability test based on a serpentine pattern is introduced. By quantifying key geometric features (filament width, height, curvature, and deflection) the test enables a reproducible and scalable comparison of inks across a wide range of formulations. This method replaces qualitative assessments with measurable, normalized metrics, providing a robust tool for ink development and printer calibration.**

## 1. Introduction

Additive manufacturing (AM) has its roots in the rapid prototyping innovations of the early 1980s. The first reported technique appeared in 1981, when 3D plastic models were produced layer-by-layer using UV-cured photopolymer resin.[1] This concept—producing objects additively rather than subtractively—initially gained little attention until the stereolithography process was invented in 1984.[2] This marked the beginning of AM technology in the industry, which at the time was mainly used for the rapid production of prototypes. Over the next decade, AM techniques diversified: fused deposition modeling—based on the extrusion of thermoplastic filament—and selective laser sintering—based on a laser to fuse powdered material—were introduced alongside several other methods.[2] During this time, the technology evolved from pure prototyping to the production of functional parts, which led to experts increasingly using the term "additive manufacturing" for these techniques: gradually, the approach of building objects layer-by-layer was recognized as a new manufacturing paradigm. An important advance in the late 1990s was the development of direct ink writing (DIW), an extrusion-based AM technique for the patterned deposition of fluid "inks".[3] In 1997, researchers at Sandia National Laboratories pioneered a DIW precursor known as robocasting.[4] In this direct-write process, a continuous filament of colloidal ink is extruded, allowing free-form shaping without the need for molds. By the mid-2000s, DIW had matured into a versatile AM method. The peculiarity of this technique is that the material to be printed is directly deposited without being thermally treated before extrusion, as in fused deposition modeling, or selectively cured, as in photocuring-based technologies. The ink itself must therefore possess self-supporting ability, that is the ability to retain its shape after being deposited, while the final consolidation of the object is usually achieved through thermal treatments. The schematic of a 3D printer for DIW, shown in **Figure 1**, is rather simple: the ink is stored in a reservoir (the barrel) and pushed out of the nozzle using a dispensing mechanism.[5] A screw, a piston or a pneumatic device can be used to dispense the ink.[6] The size of the nozzle determines the size of the filament that is deposited on the printer bed and thus

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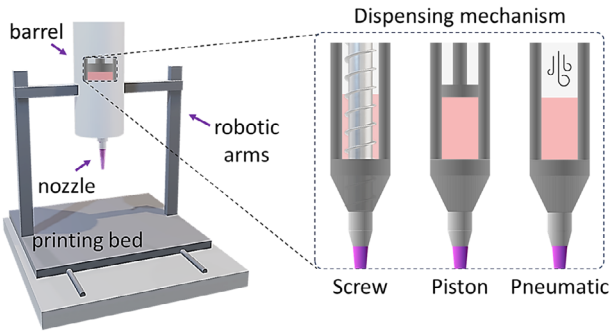


**Figure 1.** Schematic representation of a 3D printer for DIW.

the final resolution of the printed object. It is worth noting that, with the aim of ensuring proper adhesion to the printing bed and enhancing the inter-layer adhesion, the filament is usually flattened on the printing bed. To do that, the distance between the nozzle tip and the printer bed (standoff distance) is usually set to be lower than the nozzle diameter. The reciprocal motion of the bed and the head of the printer ensures the deposition of the ink in three dimensions.

Highlights

The effectiveness of printability criteria in literature is critically reviewed
Printability for Direct Ink Writing is defined as the success of five subfunctions
Dimensional analysis compares the effect of material- and process-related parameters
A standardized test is proposed for quantitative printability evaluation
Examples of trade-offs between parameters across subfunctions are shown

With respect to other AM techniques, DIW stands out for its ability to process particle-containing inks, ranging from low (2%vol./vol.) to high solids loading (typically up to 50–65%vol./vol.).[7] This allowed for the spread of DIW to manufacture wide range of parts, ranging from structural advanced ceramics,[8,9] biomaterials,[10,11] electronic devices,[12] piezoelectric advanced sensors,[13,14] and electrodes for power generation and energy storage.[15] More in general, DIW is celebrated as one of the most versatile 3D printing techniques, capable of printing very different types of materials (polymers, ceramics, metals, gels, and more)[16] as long as a suitable ink with the right rheology is provided: it must be able to flow as the load is applied and must suddenly recover the solid-like behavior after its deposition.[17] The search for this balance pushed researchers and engineers to meticulously refine ink compositions with the aim of attaining desired shear-thinning, yield stress, and flowability attributes.[18] Unfortunately, the exploration of such a wide range of ink formulations was not always accompanied by a systematic study of the phenomena determining ink printability, leaving the ink optimization phase to user experience and iterative trial-and-error procedures. Aiming to impose order on the still chaotic literature on ink printability-window, the manuscript is structured as follows.

- In section 2 we critically revise the printability criteria proposed so far in the literature, highlighting the impossibility to define universal parameters able to define printability of a given ink.
- In section 3 we redefine the concept of printability relating it to the success of five distinct subfunctions (extrudability, single filament accuracy, planar accuracy, buildability, and capability to produce suspended structure). Each subfunction has been analyzed in detail, identifying the correlations among the material- and process-related parameters with the help of dimensional analysis.
- In section 4 a quantitative way to measure the performance of inks during printing is proposed aiming to i) overcome the binary concept of printability and avoid discretion in the classification of inks; and ii) provide a tool to control the design and fine tuning of materials by relating the analysis of printing defects to the identified dimensionless numbers.

## 2. Critical Analysis of Printability Criteria Based on Rheological Thresholds

Elastic modulus and yield stress are generally considered to be the most relevant rheological properties for the purpose of printability of inks for DIW. Most of the rheological criteria proposed so far attempted to link printability to threshold values for one or both these parameters. However, the inappropriacy of the rheological methods commonly used to measure such properties resulted in system-specific and/or chaotic printability windows. Before critically reviewing the literature on rheological printability criteria, it is useful to briefly summarize the meaning and assessment methods of elasticity and yield stress of viscoelastic fluids such as the inks for DIW.

In shear mode, the elastic modulus (G) is the ratio between the shear stress ($\tau$) required to generate a non-permanent shear deformation ($\gamma$), $G = \tau/\gamma$. Regarding the yield stress ($\tau_y$), it represents the minimum stress at which the material begins to deform permanently (plastically). In shear, it is measured by subjecting the materials to increasing values of shear stress until plastic deformation. In the vast majority of cases, the rheological properties of the inks are measured by means of oscillatory shear experiments. Provided the oscillation amplitude is low enough to be in the linear viscoelastic regime, such experiments provide the frequency-dependent elastic and viscous modulus of the ink, respectively referred to as $G'(\omega)$ and $G''(\omega)$. In DIW, inks with solid-like behavior (i.e. $G' \gg G''$ and both moduli are weakly dependent on the oscillation frequency) are often preferred for the purpose of printability. For such inks, the low-frequency elastic shear modulus represents a reasonable estimate of the ink elasticity, $G = \lim_{\omega \to 0} G'(\omega)$. In many cases, however, the hypothesis of solid-like behavior cannot be invoked, and the most meaningful estimate of the ink elasticity is given by the value of $G'$ at a frequency that matches the printing conditions, i.e., close to the reciprocal of the characteristic time of the process. Surprisingly, many authors identify the ink elasticity by merely referring to the elastic modulus in the low stress limit from stress sweep experiments ($G'_0$) that are carried out at an arbitrary frequency (often ≈1 Hz). The same test is also used to infer the oscillatory yield stress ($\tau_y^*$), identified as the value of the oscillatory stress


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at which the linear viscoelastic regime is exceeded (drop of G′ or the G′–G″ crossover).[19–27] More rigorous experimental methods based on non-oscillatory, steady state flow tests have been less frequently adopted.[28–33] In such cases, $\tau_y$ is identified by fitting to the experimental data some rheological model specific for non-Newtonian fluids exhibiting yield stress. In this regard, the work by Dinkgreve et al.[34] is a must read for an in-depth understanding of yield stress evaluation by rheological approaches.

In the light of the previous considerations, it is not surprising that rheological criteria proposed in the past to define printability have led to numerical thresholds which can vary even by orders of magnitude. Another reason why it is very difficult to define universal printability thresholds is the huge difference between the rheological properties of the materials used for DIW, which vary from very soft inks to highly structured pastes. Despite these premises, reviewing in detail the printability windows in DIW proposed in the recent literature is essential to overcome the current limitations and propose a more general approach, which will be presented in Section 3.

Among the first attempts to define some rheological criterion to classify an ink as printable, it is worth citing the work by Minas et al., who dealt with particle-stabilized emulsions and foams for DIW.[35] Stress sweep tests at 1 Hz were carried out to determine $G'_0$ and $\tau_y^*$, taken at the crossover of the moduli. The qualitative analysis of the shape retainment yielded the following threshold criteria for printability:

$$\begin{cases} G'_0 > 2 \cdot 10^3 \text{ Pa} \\ \tau_y^* > 2 \cdot 10^2 \text{ Pa} \end{cases} \tag{1}$$

Working on aqueous clay paste suspensions, Chan et al. proposed a similar printability criterion based on a combination of the elastic modulus and the oscillatory yield stress measured through a strain sweep test at 10 rad $s^{-1}$.[36] Also in this case, $\tau_y^*$ was measured at the crossover of the moduli. The following relationship was derived:

$$G'_0 \cdot \tau_y^* \geq k \tag{2}$$

where $k$ is within the range of $\approx 6 \times 10^6$ to $\approx 1 \times 10^7$ $Pa^2$. In a follow-up work on colloidal particle-stabilized emulsions, the value of the $k$ parameter was further refined to be $\approx 5 \times 10^6$ $Pa^2$.[37]

Another contribution to the field is due to García-Tuñon and co-workers.[38] According to their results, collected through strain sweep tests at 0.5 Hz, ink printability can be achieved when the following criteria are satisfied:

$$\begin{cases} G'_0/\tau_y^* \geq 20 \\ \tau_y^* \geq 500 \text{ Pa} \end{cases} \tag{3}$$

Again, $\tau_y^*$ was evaluated at the crossover of the moduli. In a subsequent work, the same group reconsidered the threshold limits for the yield stress on the basis of empirical evidence.[39] On the one hand, an upper boundary was introduced to account for irregular extrusion and clogging occurring for extremely stiff inks. On the other hand, the lower boundary was widened to account for the printability of softer materials, which are typically required in the biomedical field.[40,41] This led to the following revised criteria for printability:[39]

$$\begin{cases} G'_0/\tau_y^* \geq 20 \\ 250 \leq \tau_y^* \leq 1500 \text{ Pa} \end{cases} \tag{4}$$

The specificity of the inks considered and the arbitrariness of the frequency at which the oscillatory tests were carried out makes the criteria in Equations (1–4) hardly generalizable to different systems. Their inadequacy clearly emerges from **Figure 2a**, in which rheological data collected from literature for 45 printable and non-printable formulations are superimposed to the predictions of Equations (1), (2), and (4). All data were taken from stress sweep experiments carried out at ≈1 Hz, from which $G'_0$and $\tau_y^*$ evaluated at the crossover of the moduli are considered.

None of the proposed criteria successfully distinguishes between printable and non-printable inks. In addition, there are several inks with almost identical rheological properties that were categorized differently by the authors of the respective papers. This reflects the "original sin" behind Figure 2a, namely relating the printability of inks to rheological parameters obtained from oscillatory shear tests at an arbitrary frequency. As discussed at the beginning of this section, inks for DIW are indeed viscoelastic materials, that often exhibit frequency-dependent elastic and viscous moduli (as a result, $\tau_y^*$ evaluated at the G′–G″ crossover can be frequency-dependent too). Frequency sweep tests, which are rarely reported for DIW inks,[42,43] are instead ideal for investigating how the ink behaves both under rapid deformations and at rest (conditions mimicked by high and low frequency values, respectively).[44] As shown schematically in Figure 2b, two inks can therefore behave very differently during printing, and yet have deceptively comparable values of G′ at a given test frequency, ultimately resulting in strikingly different printability outcomes. To better mimic the printing conditions, several authors considered in their printability analysis also non rheological parameters, such as the surface tension and gravity.[28,33,45–47] Even in such cases, however, the tests for the assessment of the properties of the inks did not match the flow conditions experienced by the inks during printing, making the derived rheological parameters not meaningful for the definition of general printability criteria. The inadequacy of threshold-based criteria for printability has been recently questioned also by Li et al.[16] In addition to the previous considerations, another issue underlying the difficulty of generalizing the proposed criteria concerns the definition of printability itself, since the evaluation of the printing behavior of the inks is typically not grounded on any measurable feature but merely subjected to personal assessment.

Given the complexity of the DIW printing process, the enormous variability of the rheological properties of inks, and the lack of rigorous testing protocols for the correct determination of their rheological properties, in the next section we propose a paradigm shift. Following the approach of Rau et al.,[48] the printing process is broken down into five distinct subfunctions, for each of which the material and process-related parameters responsible for printability are identified and their effect is discussed in detail.


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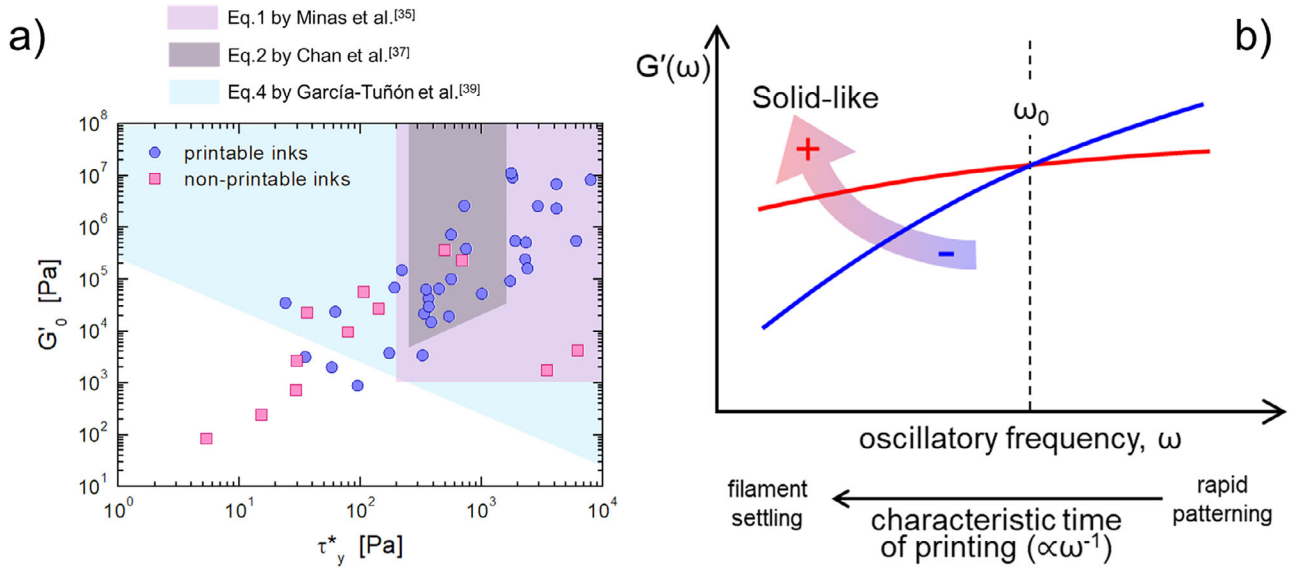


**Figure 2.** a) Comparison of literature data on rheological properties of DIW inks and printability criteria reported in Equations (1), (2), and (4). Details on the inks are collected in Table S1 (Supporting Information). b) Schematic showing the frequency-dependence of the elastic modulus of two viscoelastic inks with different degrees of solid-like behavior. Though having comparable G′ values at around a given frequency (here $\omega_0$), the elasticity of the two inks can be very different while printing, leading to different printability results. The process, indeed, spans a wide interval of characteristic times, corresponding to low (e.g., soon after filament deposition) or high frequencies (e.g., during rapid filament patterning).

## 3. Beyond Threshold Criteria: Printability as a Result of the Success of Basic Subfunctions

The journey that the ink must travel during printing can be divided into five subsequent phases, hereinafter referred to as "subfunctions":

1) extrudability;
2) single filament accuracy;
3) planar accuracy;
4) buildability;
5) capability to produce suspended structure.

In the first stages of printing, the ink must be able to flow easily enough to be extruded evenly through the nozzle ("extrudability"). Soon after exiting the nozzle, the filament must recover a solid-like behavior to resist the forces that could deform it and retain the prescribed cross-section ("single filament accuracy"). In this second phase of the printing process, the rheological features of the material must ensure that the pattern specified by the design can be accurately reproduced, which may include tight bends and/or tiny gaps between filaments ("planar pattern accuracy"). Finally, in the late stages of printing, each layer must not collapse during stacking, retaining its shape under the weight of subsequent layers ("buildability") and allowing the deposition of "suspended structures" that may be dictated by the design. It is important to observe that, in some cases, these requirements can conflict with each other. For example, good extrudability may result in scarce single filament accuracy, as a soft filament could collapse under its own weight after deposition. Similarly, an ink that is stiff enough to allow the production of suspended structures could not be flexible enough to reproduce intricate patterns, which means low planar accuracy. Overall, the success of each subfunction concurs in a general definition of "printability". Both material- and process-related parameters play a role, but their complex inter-relationships are often unknown. In this section, we collect the relationships proposed by several authors and rationalize their efforts by resorting to dimensional analysis. In particular, when analytical relationships among the variables are not available, we use the Buckingham's Π theorem[49] to identify the main dimensionless groups responsible for the success of each subfunction. The aim is not only to provide the operator with the most important parameters to control, but also to highlight the most effective ones to work with in order to optimize the different steps of the printing process.

### 3.1. Flow Inside the Nozzle

The first step in DIW is to fill the material into the reservoir and then extrude a filament through the nozzle of the printer. This seemingly simple process can be studied at different levels of detail.[50–54] The simplest approach aims to ensure a continuous and uniform flow inside the nozzle by adjusting the printing parameters, such as extrusion speed, nozzle length and pressure at nozzle inlet. A more comprehensive analysis of the problem, however, must not overlook the (often complex) rheological behavior of the material. The flow of the ink inside the barrel and/or the nozzle falls in the general category of a flow of a non-newtonian fluid inside a channel. Depending on the rheological behavior of the material, defined by the constitutive equation of the ink and on the nozzle geometry, different analytical relationships between the volumetric flow rate (Q) and the pressure at the nozzle entrance (ΔP) have been derived.[50,53–55] As for the rheology of the ink, it is generally accepted that it should have yield stress and high viscosity at low shear rates, shear thinning


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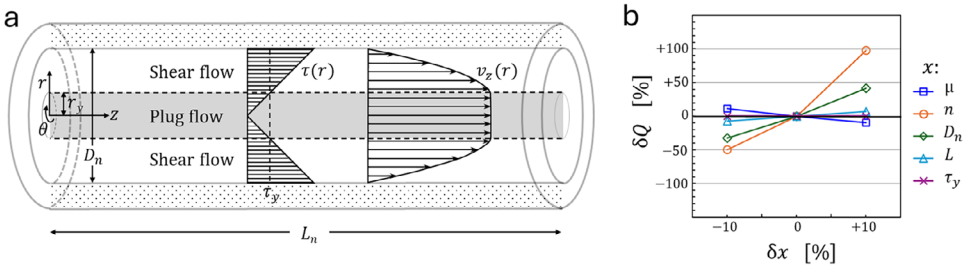


**Figure 3.** a) Shear stress and velocity profile in a Herschel-Bulkley fluid flowing in a cylindrical nozzle. b) Fluctuations of the volumetric flow rate ($\delta Q$) for a H-B fluid ($\mu = 10^2$ Pa s$^n$, $n = 0.5$, $\tau_y = 10^3$ Pa) flowing in a cylindrical die ($D_n = 2 \times 10^{-3}$ m, $L_n = 10^{-2}$ m) under a pressure drop $\Delta P = 5$ bar in response to variations of ±10% of various parameters, ($\delta x$).

behavior to facilitate extrusion through the nozzle, and quickly return to high viscosity after extrusion to ensure shape retention ability.[56] Regardless of its origin, this behavior can be satisfactorily described by the Herschel-Bulkley (H-B) model, which is versatile enough to well describe most of the inks used for DIW:[6,28,33,57]

$$-\mu\dot{\gamma}^n = \begin{cases} 0 & if\ \tau < \tau_y \\ \tau - \tau_y & if\ \tau \geq \tau_y \end{cases} \tag{5}$$

In Equation (5), $\mu$ is the consistency index of dimension, $\dot{\gamma}$ is the shear rate, n is the flow index ($0 < n < 1$ for a shear-thinning fluid), $\tau$ is the shear stress, and $\tau_y$ is the yield shear stress. Equation (5) states that the layers of the ink do not move or move at the same rate ("plug flow") in those regions of the channel where the stress is below $\tau_y$, while it flows as a power-law fluid where $\tau_y$ is exceeded. Excluding the limiting case of absence of flow (i.e., $\tau < \tau_y$ wherever in the channel), the velocity profile of an H-B ink in a cylindrical nozzle with length $L_n$ and diameter $D_n$ is shown in **Figure 3a**.

Assuming the absence of wall slip phenomena and integrating the velocity profile on the nozzle section, one gets the volumetric flow rate inside the nozzle:

$$Q = \frac{8\pi\left(\frac{\Delta P}{L_n}\right)^n}{\mu\,(n+3)}\left(\frac{D_n}{4}\right)^{n+3} \cdot g\left(r_y, n\right) \tag{6}$$

where $g(r_y,n)$ is a function of $r_y$ and n (for an analytical expression of $g(r_y,n)$, see Ref. [55]).

As far as the ink extrudability is concerned, the challenge is to finely control the volumetric flow rate at the nozzle exit. Fluctuations in the flow rate would in fact prevent the filament width and height specified by the design from being achieved, which would compromise the quality of the subsequent phases and, ultimately, of the printed object.

Depending on the dispensing mechanism (see the Introduction), printers for DIW can be divided into two categories: i) flow rate-controlled (e.g., those equipped with an extruder); and ii) pressure-controlled (e.g., those equipped with a pneumatic dispensing mechanism). For the first category, assuming the incompressibility of the ink, the steadiness of Q and, hence, of the cross-section of the filament at the nozzle exit, will reflect the accuracy of the pumping system: the more continuous and precise is the Q(t), the more regular will the section of the extruded filament throughout the process. For pressure-controlled printers, where the uniformity of the filament depends on the stability over time of each of the parameters mentioned in Equation (6), it is more difficult to guarantee a uniform Q. Once the temperature is set, $\tau_y$, $\mu$ and n are in principle fixed and remain constant over time. Nevertheless, fluctuations in the rheological parameters can occur due to the inherently complex rheology of the fluid. Common ink formulations, for example, may contain highly anisotropic particles that can gradually align during flow. Another source of fluctuation reflects the uneven distribution of the components (solid particles, plasticizers, fluid phases) present in real inks/pastes. Ensuring a good mixing of the components before printing is, hence, crucial to make the rheological parameters uniform in space, minimizing irregularities in the filament section. Material inhomogeneity can also indirectly affect process-related parameters. For example, a progressive accumulation of solid particles within the nozzle or at its entrance/exit may indeed alter the effective values of $D_n$ and $L_n$ in Equation (6).

Figure 3b shows the sensitivity of the volumetric flow rate to variations of ±10% of the "x" parameters of Equation (6) for a generic H-B fluid ($\mu = 10^2$ Pa s$^n$, $n = 0.5$, $\tau_y = 10^3$ Pa) flowing in a cylindrical die ($D_n = 2 \times 10^{-3}$ m, $L_n = 10^{-2}$ m) under a pressure drop $\Delta P = 5$ bar. Notably, substantial fluctuations in the flow rate are expected in response to changes in $n$, with $Q$ that almost doubles when $n$ increases by only 10%. Appreciable fluctuations (up to ≈40%) are predicted in response to changes in the effective nozzle diameter $D_n$. Less relevant is the effect of changes in $\mu$, $L_n$ or $\tau_y$.

Before concluding this section, we observe that the previous discussion neglects wall slip phenomena. Such an assumption, which is generally fulfilled at the typically moderate flow rates of DIW, can break down in case of highly structured fluids (e.g. ceramic pastes or polymers) or inhomogeneous suspensions of solid particles.[58,59] For extrudability purposes, wall slip could destabilize the flow causing an unsteady discharge of material. Different strategies can be pursued to minimize wall slip such as acting on the nozzle (e.g., making it rougher or treating it chemically to improve adhesivity with the ink), changing the printing conditions (e.g., increasing the average pressure in the nozzle to enhance nozzle-ink adhesion), or modifying the ink formulation (e.g., reducing the amount of lubricating components). Reviewing these strategies goes beyond our current purposes. Here we simply recommend to account for possible wall slip phenomena when interested in the extrudability of an ink, referring to specific papers for an in-depth analysis of the phenomenon.[58,60,61]


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**Table 1.** Description of the flow inside the nozzle: equation governing the phenomenon, sources of the defects and possible remediation strategies.

| | Main goal | Assumptions | Parameters |
|---|---|---|---|
| | Steadiness/control of the volume flow rate | Herschel-Bulkley fluid: $-\mu\dot{\gamma}^n = \begin{cases} 0 & \text{if } \tau < \tau_y \\ \tau - \tau_y & \text{if } \tau \geq \tau_y \end{cases}$ Incompressibility Cylindrical nozzle (diameter $D_n$, length $L$) | $\mu$: consistency index; $n$: flow index; $\tau_y$: yield shear stress |
| Q-controlled printers | **Governing equation** | $Q$=cost | |
| | **Main source(s) of defect** | **Possible action(s)** | |
| | Inaccuracy of the pumping system | Improving the reliability of the pumping system | |
| $\Delta P$-controlled printers | **Governing equation** | $Q = \frac{8\pi\left(\frac{\Delta P}{L_n}\right)^n}{\mu(n+3)}\left(\frac{D_n}{4}\right)^{n+3} \cdot g(r_y, n)$ | **Definitions**: $\Delta P$: pressure drop along the nozzle; $g(r_y, n)$: analytical function (see [35]) |
| | **Main source(s) of defect** | **Possible action(s)** | **Note(s)** |
| | Fluctuations of the rheological parameters | Stabilize the temperature, homogenize the ink, stabilize the particles (if any) | High sensitivity of $Q$ to changes in the value of $n$; smaller effect in response to changes in $\mu$ and $\tau_y$ |
| | Inadequate mixing of the constituents of the ink | Homogenize the ink, stabilize the particles (if any), match the densities of the ink's constituents, viscosize the ink to slow down sedimentation/buoyancy phenomena, … | |
| | Wall slip | Use rough nozzles, use nozzles with bigger diameter | |

**Table 1** resumes the main highlights of the "flow inside the nozzle" step.

### 3.2. Flow Outside the Nozzle

#### *3.2.1. Single Filament Accuracy (SFA)*

Once the material has been extruded and is out from the nozzle, a filament is formed and deposited on the previous layer (or, in the case of the first layer, on the printing plate). This step is crucial for the overall printing quality. The size and shape of the deposited filament reflect the complex interplay between many material- and process-related parameters. Let us first analyze the filament when it has just come out of the nozzle without having yet touched the printing plate. For viscoelastic inks (e.g. polymers or polymer-containing inks/pastes), the cross-section ($S_f$) can be slightly larger than that of the nozzle ($S_n$) because of "die swell". This phenomenon consists in the swelling of the extruded filament at the nozzle exit, and it is due to the sudden relaxation of the normal stresses accumulated in the material when converging into the nozzle at its entrance (**Figure 4**).[62,63] To reduce die swell, the residence time of the ink inside the nozzle must be prolonged to allow the relaxation of the normal stresses. This can be achieved by reducing the extrusion rate or by using longer nozzles. The use of tapered dies (e.g. conical nozzles) is also recommended, as the gradual convergence of the flow at the nozzle entrance minimizes the accumulation of normal stresses in the material, resulting in a reduction of die swell. Besides affecting the cross-section of the filament, the complex fluid dynamic at

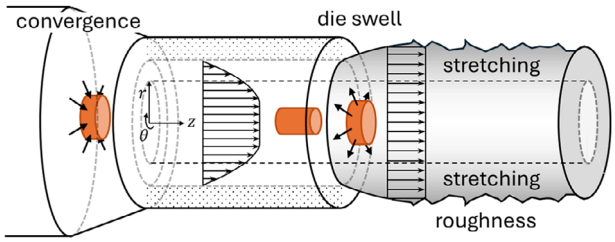


**Figure 4.** Velocity profiles inside and outside a cylindrical nozzle. The swelling of a small volume of material when exiting from the nozzle is highlighted. The rearrangement of the velocity profile when due to the sudden change in the boundary condition causes the stretching of the outer shell of the filament at the nozzle exit. This can damage the surface of the filament, making it rough.


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the nozzle exit also affects its surface texture. In fact, the abrupt change of the boundary conditions at the nozzle exit causes a severe stretching of the outer shell of the filament. If this is not sufficiently compliant, surface damage can occur, making the filament rough (Figure 4). To be noticed that, if the ink does not show a marked viscoelastic behavior, die swell is a limited phenomenon and therefore can be neglected. To improve the surface texture of the filament, one can reduce stress at the nozzle exit (e.g. by using bigger nozzles and/or reducing the extrusion rates). Another possible action could consist in promoting wall slip to reduce the tear stresses at the nozzle exit responsible for a rough surface texture of the filament. Possible strategies envisage acting on the nozzle (e.g. polishing its inner surface or using non-stick coatings), printing conditions (e.g. decreasing the average pressure in the nozzle to reduce nozzle-ink adhesion), and/or ink formulation (e.g. by adding slip-inducing additives).[58,60,61] We observe that promoting wall slip is in contrast with what emerged when addressing extrudability issues (see Section 3.1). The need to conciliate opposite requirements is not surprising and reflects the intrinsic complexity of the 3D printing process.

Let us now consider the phase in which the filament is deposited on the printing plate or on the previous layer. We will first consider only the deposition of a straight filament at the bottom, focusing on the size and shape of its cross-section. Adhesion to the underlying material (either the printing plate or a previous layer) anchors the filament, which can lead to bulging/bending depending on whether the relative velocity between the nozzle and the adhesion layer ($v_n$) is higher or lower than the average velocity of the material at the nozzle exit ($\bar{v} = Q/S_n$). An "ink-to-nozzle speed ratio" (SR) can be defined as $SR = v_n/\bar{v}$ ($= \frac{v_n \pi D_n^2}{4Q}$ for cylindrical nozzle with diameter $D_n$). DIW is typically carried out at SR ≈1. Printing at SR > 1 will result in a $S_f$ that is smaller than $S_n$. In such condition of "under-extrusion" the filament may break if its cohesive strength is not sufficient to resist stretching. If, instead, SR < 1 ("over-extrusion"), then $S_f$ will be greater than $S_n$ and the filament will bulge/bend (**Figure 5**).[53,64–66]

Once the filament has been deposited on the printing plate or on the previous layer, it should accurately reproduce the cross-section imposed by the design. Depositing a flat trace, rather than a rounded one, improves adhesion between layers, reduces voids between adjacent filaments and ensures a smoother surface texture of the printed object.[4,44,52,67] To promote a flattened shape, it is possible to act on the geometry of the nozzle or reduce its distance from the previous layer ("standoff distance") to squeeze the filament (**Figure 6a**). However, preserving the cross-section is anything but easy. In fact, the deposited filament is subjected to several forces (e.g. gravity, surface tension, interaction with the underlying material) that seek to deform it, and the ability to resist these forces depends on the yield strength of the material. Single filament accuracy can be defined as the ability to faithfully reproduce the height (H) and width (W) specified by the design. As such, the SFA will depend on the balance among all the forces involved. Such a general principle has been mathematically formalized in different ways for inks made of carbopol gels,[33] alumina,[45] and printable concretes:[47]

$$\tau_y \cdot \sqrt{S_f}/\Gamma \gg 1 \quad (7)$$

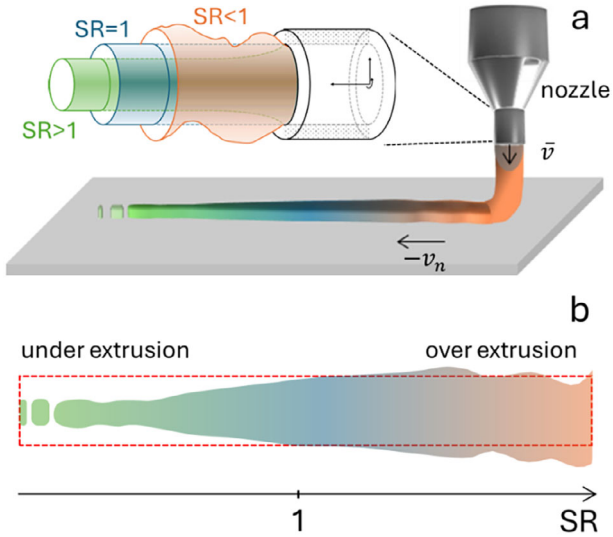


**Figure 5.** Schematic showing the deposition of a straight filament by DIW onto a substrate. Colors indicate print defects as a function of ink-to-nozzle speed ratio: under extrusion with filament breaking (SR > 1; green); steady extrusion (SR ≈1; blue); over extrusion (SR < 1; orange). b) Top view of the filament. The red dashed line schematizes the area in which the filament would fit in case of SR = 1.

$$\tau_y^* \geq \Delta P_L \quad (8)$$

$$G'_0 \geq \rho g H/\epsilon_{tol} \quad (9)$$

In Equation (7), $\tau_y$ is the yield stress evaluated by fitting the flow curves to the Herschel-Bulkley model, Γ is the surface tension of the ink material, and $S_f$ is the filament section area; in Equation (8), $\tau_y^*$ is the yield stress from a fixed-frequency oscillatory stress sweep test (here taken at the interception of the tangent to the linear viscoelastic region with the tangent of the final part of the G′ curve) and $\Delta P_L$ is the Laplace capillary pressure; in Equation (9), $G'_0$ is the elastic shear modulus evaluated from a stress sweep test in the linear viscoelastic region (at a given oscillatory frequency), g is the gravitational acceleration, and $\epsilon_{tol}$ represents the maximum admissible tolerance in the filament deformation (operator-defined parameter). The threshold values to be exceeded in the inequalities 7–9 depends on the specific case. For low viscosity inks, it would be sufficient that the yield strength

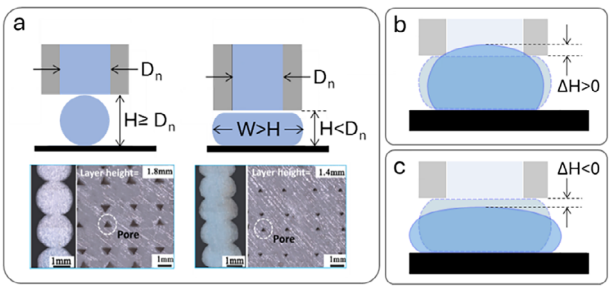


**Figure 6.** a) Effect of the standoff distance on the overall structure porosity. Micrographs highlight the inter-filament porosity. Readapted from.[67] Increase b) and decrease c) of the filament height.


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offsets the surface forces ($\approx\Gamma$ or $\approx\Delta P_L$) that cause the increase of H of the filament (Figure 6b). Differently, in case of concrete filaments,[47] the gravitational term ($\approx\rho gh$) plays a major role and the challenge is limiting slumping (decrease of H; Figure 6c). It should be noted that rheological properties in Equations (8), (9) can be frequency-dependent and thus their estimation at an arbitrary fixed frequency can be not meaningful (see Section 2). Moreover, all the references cited refer only to the surface tension of the material, whereas a more rigorous approach would distinguish between the deposition of the first layer, in which the wettability of the printing plate determines the spread of the filament, and the deposition of subsequent layers, in which the surface tension is the determining force. Concerning the first layer, the surface energy-related forces also depend on the material of the printing plate, and either spreading or rounding of the filament section can be promoted depending on the value of the spreading coefficient (S), defined as the difference between the ink/plate work of adhesion and the work of cohesion of the ink. For subsequent layers, instead, the material is deposited onto itself. In this case, the surface tension of the ink tends to lower the filament height.

Instead of looking for system-specific criteria, we resort here to dimensional analysis, a technique able to simplify complex physical problems by reducing the number of variables involved. According to the Buckingham $\Pi$ theorem,[49] the m variables a physical problem depends on can be grouped into $\Pi$ = m-k dimensionless groups, k being the number of fundamental dimensions. For purely mechanical problems, such as those we are interested in, the fundamental dimensions are mass, length and time, hence k = 3. The aim here is not only to reduce the number of variables, but also to identify dimensionless groups in order to be able to easily compare the relative importance of the physical quantities involved. Regarding the SFA, the problem consists in finding a relationship between the deviation from the design-imposed height ($\Delta$H) and the material- and process-related parameters involved, which are the stand-off distance (H), the yield stress ($\tau_y$), the density ($\rho$), the gravitational acceleration (g) and the surface tension ($\Gamma$) or the spreading coefficient of the ink (S), depending on the underlying layer on which the filament is deposited. Since k = 3, these m = 6 dimensional variables can be grouped in 3 dimensionless groups, and the SFA problem can be formulated as follows:

$$\frac{|\Delta H|}{H} = \Phi_1\left(\frac{\tau_y}{\Gamma/H}; \frac{\tau_y}{\rho g H}\right) \tag{10a}$$

$$\frac{|\Delta H|}{H} = \Phi_1\left(\frac{\tau_y}{S/H}; \frac{\tau_y}{\rho g H}\right) \tag{10b}$$

where $\Phi_1$ is a generic function and $\frac{|\Delta H|}{H}$, $\frac{\tau_y}{\Gamma/H}$, (or $\frac{\tau_y}{S/H}$) and $\frac{\tau_y}{\rho gH}$ are a possible set of dimensionless groups which govern the physics of the problem. In detail, $\frac{|\Delta H|}{H}$ is a measure of the SFA, which depends on $\frac{\tau_y}{\Gamma/H}$ (or $\frac{\tau_y}{S/H}$) and $\frac{\tau_y}{\rho gH}$. These three groups compare the ability of the yield stress of the material to contrast i) the surface tension; ii) the building plate wettability; and iii) the inertial mass of the filament. Assessing the analytical expression of $\Phi_1$ to define quantitative threshold criteria goes beyond our purposes. Here we only aim at highlighting that, to improve the SFA (i.e., to minimize $\frac{|\Delta H|}{H}$) one can act on the material (e.g. by tuning $\Gamma$ or S, increasing $\tau_y$, decreasing $\rho$) or changing the printing parameters (e.g. the stand-off distance, the ink-to-nozzle speed ratio) to reduce H. It is important to observe that dimensional analysis does not provide a unique combination of dimensionless groups. Another possible dimensionless group relevant to the SFA is $\frac{\rho gH}{\Gamma/H}$, a.k.a. the Eotvos (Eo) or Bond number (Bo), which compares gravitational force and surface tension. The relevance of the Eo was already highlighted by Alvarez and co-workers.[68] From a practical point of view, by estimating the Eo number, one can understand which prevails between surface tension and gravitational force for the specific ink one is dealing with. Subsequently, the dominant force that compromises the SFA can be compared to the resisting force due to the yield stress. In this perspective, Equation (7) is more suited for a soft carbopol gel, in which the surface tension plays a major role (i.e., Eo is small). Differently, Equation (9) better applies to a printable concrete, for which Eo is big and the SFA mostly depends on the ability of the filament not to collapse under its own weight. Generalizing the concept, the dimensionless groups relevant to the SFA express the ratio between three parameters having the dimensions of a stress: $\rho$gH, $\frac{\Gamma}{H}$, and $\tau_y$. By comparing these values in pairs, the SFA of a particular ink can be optimized by acting either on the material- ($\rho$, $\Gamma$ or S, and $\tau_y$) or on the process-related parameters (H) (**Table 2**).

Finally, we add a few considerations on the way in which $\Gamma$ and $\tau_y$ are defined and how they are experimentally assessed. Regarding the surface tension, for solvent-based inks some authors assume $\Gamma$ is equal to those of the solvent,[28] but the addition of particles can remarkably affect its value. As an example, the addition of a 0.3 vol.% of graphene oxide to water has been found to cause a drop in surface tension from 72 to 30 mN m$^{-1}$.[38] Therefore, for the purpose of assessing the SFA, it is recommended to always refer to the $\Gamma$ of the complete formulation of the ink. Regarding $\tau_y$, it is measured through a steady state flow test from low to high stress and fitting the Herschel–Bulkley model to the experimental data. Rheological tests based on oscillatory measurements at increasing oscillation amplitude (e.g. strain/stress sweep experiments) are generally easier to perform but the outcomes of this kind of dynamic tests (e.g., some threshold value of G′ or of the oscillatory shear stress) are not necessarily correlated with the $\tau_y$ assessed by steady state tests. Therefore, it is recommended to adhere to the steady state flow tests as the most rigorous procedure for measuring $\tau_y$. Notably, two different types of yield stress can be defined for most of the inks for DIW, and they can be easily distinguished through a stress–shear rate loop protocol, as highlighted by M'Barki et al.:[28] a static ($\tau_y^{stat}$) and dynamic yield stress ($\tau_y^{dyn}$), which respectively correspond to the stress required to flow from a rest state and the minimum stress required for a fluid in motion to continue flowing. While the former is more idoneous to characterize the extrudability phase, $\tau_y^{dyn}$ is instead the correct parameter to consider for SFA.

### 3.2.2. Planar Accuracy (PA)

Besides matching the cross-section imposed by design, the filament must be able to accurately reproduce even intricate patterns on a single plane. The single layer may consider i) very close yet distinct filaments and ii) sharp angles. Regarding the first point,


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**Table 2.** Description of the flow outside the nozzle. Printing subfunction are summarized together with sources of the defects, possible remediation strategies and performance indicators. The definitions of the symbols are provided in the Notation section.

| Subfunction | | Defect | Relevant parameter(s) | | | Possible action(s) | | Performance indicator |
|---|---|---|---|---|---|---|---|---|
| | | | Material-related | Process-related | Dimensionless group(s) (if any) | on the material | on the process | |
| Single Filament Accuracy (SFA) | before touching the substrate/bottom layer | Die swell | Elasticity | Residence time inside the nozzle and nozzle geometry | | Reduce the ink's elasticity through rheological modifiers | Increase the residence time by slowing down the flow rate or by increasing the nozzle length; use converging nozzles | $\frac{\lvert\Delta W\rvert}{W}, \frac{\lvert\Delta H\rvert}{H}$ should be low |
| | | Surface roughness | Compliance, stickiness | Material and surface texture of the nozzle | | Use additives to make the ink more compliant (without loss in cohesiveness) and to reduce its stickiness to the nozzle wall | Reduce shear stress at the nozzle wall by using larger nozzles, printing at lower speed, lubricating and/or modifying the nozzle wall | |
| | after being deposited | Under/over extrusion | None | Ink-to-nozzle speed ratio | For cylindrical nozzle: $\frac{v_n \pi D_n^2}{4Q}$ should be ~1 | None | Act on the nozzle speed and/or the volume flow rate to make the ink-to-nozzle speed ratio close to 1 | |
| | | Discrepancy with respect to the cross-section imposed by design | Yield shear stress, surface tension or spreading coefficient, density | Stand-off distance | $\frac{\tau_y}{\gamma/H}$ or $\frac{\tau_y}{\Gamma/H}; \frac{\tau_y}{\rho g H}$ should be high | Optimize the ink formulation to (i) increase the yield shear stress and/or (ii) to reduce surface tension/spreading coefficient and density | | |
| Planar Accuracy (PA) | | Incapability to reproduce sharp corners | Elastic shear modulus, surface tension or spreading coefficient, density | Diameter of the nozzle, velocity of the nozzle, friction with the substrate (in case of deposition of the first layer) | $\frac{\rho g D_n}{G'}, \frac{\rho g D_n}{\gamma/D_n}, \frac{\rho g D_n}{\rho v_n^2}$ should be high | Optimize the ink formulation to (i) increase the density and/or (ii) to decrease the elastic shear modulus (to be measured at a frequency $\omega \sim v_n/r_{c\,min}$) and surface tension/spreading coefficient | | $\frac{r_{c\,min}}{D_n}$ should be low |
| | | Incapability to print very close yet distinct filaments | Yield shear stress, surface tension or spreading coefficient, density | Stand-off distance | $\frac{\tau_y}{\gamma/H}$ or $\frac{\tau_y}{\Gamma/H}; \frac{\tau_y}{\rho g H}$ should be high | Optimize the ink formulation to (i) increase the yield shear stress and/or (ii) reduce surface tension/spreading coefficient and density | Decrease the stand-off distance | $\frac{\lvert\Delta W\rvert}{W}$ should be low |
| Buildability (B) | | Collapse of superimposed layers | Yield shear stress, surface tension, density | Diameter of the nozzle | $\frac{\tau_y}{\gamma/r_c}; \frac{\tau_y}{\rho g H_{tot}}$ should be high | Optimize the ink formulation to (i) increase the yield shear stress and/or (ii) reduce surface tension and density | Decrease the diameter of the nozzle | $-\frac{\Delta H_{tot}}{H_{tot}}$ should be low |
| Suspended structures (SS) | | Unable to print suspended structures between supports at a distance d | | Governing equation (for cylindrical nozzle): $\delta_{max} \cong \frac{0.1\rho g d^4}{G(1+\nu)D_n^2}$ should be low | | Optimize the ink formulation aiming at (i) increasing the low-frequency elastic shear modulus and/or (ii) reducing the density | Use nozzles with bigger diameter (N.B. quadratic dependence on D) | $\delta_{max}$ should be low |

it essentially requires that the deposited filament maintains its prescribed width W. As such, it falls into the issue of SFA. Differently, the ability to deposit the filament along a path with sudden changes in direction deserves a focused analysis.

It has been reported that a simple tuning of the printing parameters is sometimes sufficient in improving the overall planar shape fidelity.[69] A low nozzle velocity is preferable to minimize inertial forces, helping in producing narrow details with high accuracy. However, when the layer shape contains sharp corners, the partial overlapping of the nozzle path at each angle edge could result in a localized increase in the height of the layer. Raising the translational speed can help to minimize such effect by exploiting the resulting under-extrusion. Thinner filaments obtained by using small nozzles make it easier to reproduce intricate patterns. Regarding the material-related parameters, a high surface tension can result in the blunting of spiky patterns.[33,46] Similarly, excessive filament elasticity makes it difficult to impart tight curves to the filament. The possibility of printing sharp corners is instead favored by the inertial mass of the filament, which dictates how firmly it adheres to the underlying layer/printing plate by friction.

The previous concepts can be rationalized by looking for a relationship between PA, definable in terms of the minimum radius of curvature that the filament can sustain ($r_{c\,min}$), and the involved parameters, which are $D_n$, $v_n$, $G'$, $\Gamma$, $\rho$ and g. According to the Buckingham $\Pi$ theorem, we define $\Pi = 4$ dimensionless groups relevant to the PA, obtaining the following relationship:

$$\frac{r_{c\,min}}{D_n} = \Phi_2\left(\frac{\rho g D_n}{G'}; \frac{\rho g D_n}{\gamma/D_n}; \frac{\rho g D_n}{\rho v_n^2}\right) \tag{11}$$

where $\Phi_2$ is a generic function and $\frac{r_{cmin}}{D_n}$, $\frac{\rho g D_n}{G'}$, $\frac{\rho g D_n}{\gamma/D_n}$ and $\frac{\rho g D_n}{\rho v_n^2}$ are a possible set of dimensionless groups for describing the problem. $\frac{r_{cmin}}{D_n}$ is a measure of the PA, which depends on $\frac{\rho g D_n}{G'}$, $\frac{\rho g D_n}{\gamma/D_n}$ and $\frac{\rho g D_n}{\rho v_n^2}$. In detail, the ability to bend the filament forming a sharp corner essentially depends on the weight of the filament, which dictates its friction to the underlying material necessary for contrasting the deforming action of the filament elasticity, the surface tension, and the kinetic energy of the nozzle (see **Figure 7**).

In case of deposition of the first layer on a substrate, the friction coefficient is another parameter that should be enhanced. The latter "inertial effect" is essentially negligible if the standoff distance is comparable to or lower than the nozzle diameter, which means that the material is gently squeezed on the underlying layer. Apart from some research papers investigating the effect of the standoff distance on printing quality,[70] the use of a distance higher than the nozzle diameter is quite rarely observed.

To sum up, to control the PA one can act either on the material ($\rho$, $\Gamma$, and $G'$) or on the printing parameters ($D_n$ and $v_n$).

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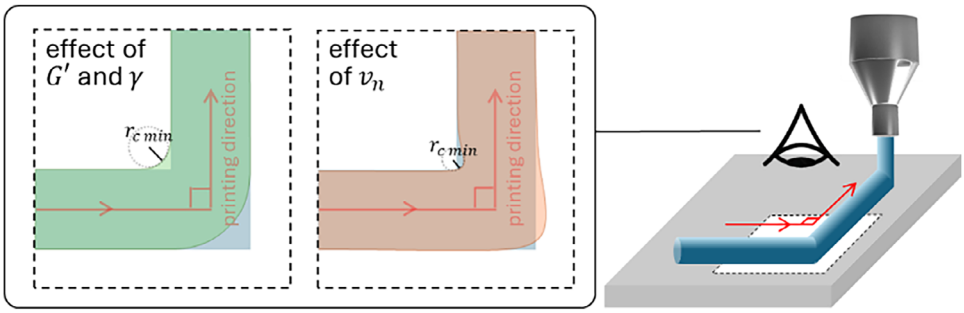


**Figure 7.** Effect of filament elasticity, surface tension and nozzle velocity on the planar accuracy. The schematic refers to the printing of a sharp right angle. Note that the defect due to an excessive $v_n$ is not observed when the standoff distance of the nozzle is comparable or lower than the nozzle diameter.

In Table 2 we summarize the expected qualitative dependence of PA on each one of these parameters. Regarding the way of measuring the parameters, the only material-related property that deserves clarification is the filament elasticity. It can be measured by means of small-amplitude oscillatory shear tests. For the purpose of PA, one should refer to G′ at a physically meaningful frequency that matches the characteristic time of directional change during printing, $\omega \sim v_n/r_{cmin}$. Since G′ increases with frequency, printing at a low nozzle velocity allows to deal with a filament that is less elastic and easier to bend, with a beneficial effect on PA.

### 3.2.3. Buildability (B)

After a single layer is complete, the "journey" of the filament for DIW continues by superimposing another layer of material. During the stacking of successive extruded layers, it is essential to ensure that the deposition of the upper layer does not affect the stability of the lower ones. Typically, a collapse of the structure can occur due to the overloading of the new layers on the still "fresh" ones. To determine whether a material is suitable to be stacked layer by layer, a criterion to assess the ink buildability must be introduced. Some authors quantified this parameter by depositing a prescribed number of layers and evaluating the mismatch between the height of the printed structure and the designed one.[28,47,71] M'Barki et al. proposed a quantitative model accounting for both the capillarity forces among superimposed 3D printed filaments (≈Γ) and the gravitational force acting on the printed structure.[28] To avoid slumping, hence, $\tau_y^{dyn}$ was proposed to counteract these two contributions:

$$\tau_y^{dyn} \geq \gamma D_n^{-1} + \rho g H_{tot} \tag{12}$$

where $H_{tot}$ is the total height of the printed structure and $D_n$ is the nozzle diameter. Although the inequality expressed in (12) well captures the physics of buildability, we believe that the most reasonable characteristic length to be considered to quantify the contribution of the surface tension is the radius of curvature of the neck that forms between two superimposed layers ($r_c$; see **Figure 8**). The action of the surface tension adds to that of the gravitational force, also promoting the slumping of the stack. The smaller $r_c$ is, the stronger this thermodynamic push is. To get an estimate of $r_c$, we notice the geometric similarity between the overlapping filaments in Figure 8 and the sintering of parallel cylinders. In the early stages of the latter process, $r_c$ is of order of half the radius of the cylinders.[72] Going back to the critical analysis of the inequality (12), regarding the yield stress that opposes slumping, the static yield stress $\tau_y$ should be considered instead of the dynamic one. To address the buildability issue in a more general way, we treat B the same way as the SFA, expressing it in terms of deviation from the sum of the heights of the superimposed filaments, $\Delta H_{tot}$ (see Figure 8). Therefore, the following generic relationship is derived:

$$-\frac{\Delta H_{tot}}{H_{tot}} = \Phi_3\left(\frac{\tau_y}{\Gamma/r_c}; \frac{\tau_y}{\rho g H_{tot}}\right) \tag{13}$$

where $\Phi_3$ is a generic function and $-\frac{\Delta H_{tot}}{H_{tot}}$, $\frac{\tau_y}{\Gamma/r_c}$ and $\frac{\tau_y}{\rho g H_{tot}}$ are a possible set of dimensionless groups that govern the problem. In particular, $-\frac{\Delta H_{tot}}{H_{tot}}$ is a measure of B that depends on how much the yield stress can contrast interfacial tension and filament weight, which causes a slumping of the stack. Grouping the parameters differently makes it possible to define other dimensionless groups. In particular, $\frac{\Gamma/r_c}{\rho g H_{tot}}$ compares gravitational force and surface tension. Since both these terms pull in the same direction, rather than providing information on B, their ratio re-

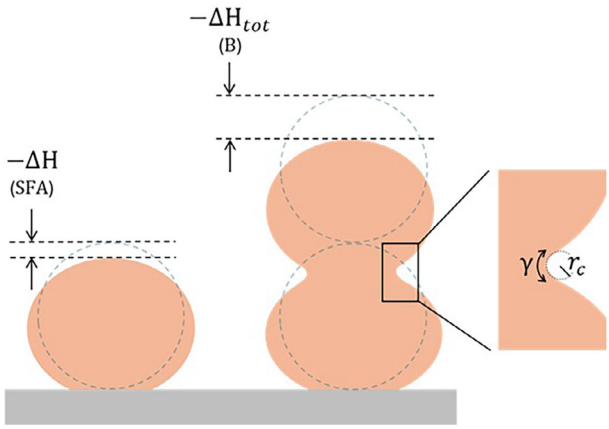


**Figure 8.** Left: Variation of the filament height after deposition of the first layer (measure of SFA). Right: Variation of the structure height after deposition of two layers (measure of B).


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veals whether one of the two plays a dominant role, possibly justifying the neglection of the other one in the analyses. It is important to observe that the lower layer may harden during the time required to stack the upper one. If this happens, then the material-related parameters in the relationship (13) should be assessed after letting the material ripen for the time needed to superimpose the upper layer.

#### 3.2.4. Suspended Structures (SS)

The last feature that a material for DIW must possess to be considered printable is the ability to build with it overhang or suspended structures in the absence of supports. Unlike the previous subfunctions defined for describing the flow outside the nozzle, for which a dimensional analysis was used to rationalize the mutual relationships among the involved parameters or groups of parameters, the capacity to build SS can be treated by referring to the classical beam theory, for which analytical relationships among the variables are available. Using such an approach, Smay et al. derived the following threshold criterion for the capacity to build SS:[46,73]

$$G' \geq 1.4\, \rho g\, s^4 D_n \tag{14}$$

where G′ is the elastic shear modulus at 1 Hz and s is the dimensionless span, defined as the ratio between the distance of the supports the filament is leaning on and the nozzle diameter. Although the inequality expressed in (14) has been effectively used as design guideline (see for instance),[74] it is based on the arbitrary assumption that the maximum tolerable deflection is 5% of the nozzle diameter. Equally arbitrary is the choice of referring to the elastic shear modulus at 1 Hz, as well as the assumption of Poisson's coefficient (ratio of transverse strain to axial strain when a material is subjected to uniaxial stress) $\nu = 0.5$ hidden in the coefficient 1.4 of (14). More in general, according to the beam theory, the maximum deflection of the filament at the midspan in the small strain limit is:[75]

$$\delta_{max} = \frac{5\rho g S_f d^4}{384 EI} \tag{15}$$

where $S_f$ is the cross-section, d is the distance between the supports, E is the Young's modulus of the filament, and I is moment of inertia of the cross-section. For a filament with circular cross-section (diameter $\approx D_n$, $I = \pi D_n^4/64$), assuming homogeneity and isotropy ($E = 2G(1+\nu)$), we can rewrite Equation (15) as:

$$\delta_{max} \cong \frac{0.1\rho g d^4}{G\,(1+\nu)\, D_n^2} \tag{16}$$

where G is the static shear modulus (ratio of shear stress to shear strain in the linear elastic region of the material's response). For viscoelastic inks, G can be replaced by the elastic shear modulus G′ in the low frequency limit (<<1 Hz), which can be easily measured through a small amplitude oscillatory shear test. It is worth noting that variations of several orders of magnitude can be recorded by changing the frequency at which the test is performed.[76] Regarding $\nu$, for non-auxetic isotropic materials, it ranges between 0 (no transversal contraction upon traction) and 0.5 (limit of incompressibility). As far as inks for DIW are concerned, $\nu$ approaches 0.5 for highly hydrated ceramic pastes/suspensions,[77] hydrogels/bioinks,[78,79] and elastomers.[80] However, much lower values have also been reported for less hydrated cement pastes ($\nu = 0.06/0.36$)[81] and thermoplastic or thermoset polymers ($\nu = 0.3 \div 0.4$).[82] Poisson's coefficients as low as 0.1 are possible for foams/pre-foamed inks.[83]

## 4. Quantitative Assessment of Printing Performance

In the previous sections, we have explained the challenges that need to be overcome in order to produce objects with good printing quality. The analysis of dimensionless numbers showed that printability relies on both materials- and process-related parameters, thus both should be controlled to ensure good printing quality. A review of the literature also revealed that there is no generally accepted quantitative method for evaluating the quality of ink and the effectiveness of the printing process. Some authors followed standard procedures to distinguish printable from non-printable inks, such as measuring the deflection of a spanning rod, evaluating the circularity of a printed shape, or assessing the final height of printed structures compared to the desired one.[6,28,73] However, in many other cases, the authors do not openly mention how they assessed the match between the desired and achieved shape, suggesting that their assessment is based on visual inspection rather than objective measurements. This makes it more difficult to compare different results from different sources. Recently, a novel approach has been proposed by Weeks et al., that smartly combined rheology, automated image analysis and machine learning to relate the shape fidelity of an ink to its behavioral features.[84] While the work of Weeks et al. is ultimately oriented at building printers able to autonomously design inks and tuning printing parameters on the fly, the approach hereafter proposed aims at providing a simple yet effective testing protocol to quantitatively assess the printability by measuring the geometric parameters introduced in the discussion of the subfunctions. The test begins by printing a single-layer serpentine shape with fixed printing parameters (printing speed, pressure, stand-off distance and so on) to measure the retainment ability of the ink (**Figure 9**).

By measuring the geometrical features of the serpentine, the simultaneous analysis of extrudability, single filament, and planar accuracy is performed. In the second step, a multi-layer serpentine is then printed to assess buildability. Finally, a bending element can be printed on top to measure the deflection of a suspended truss.

After printing, accurate image acquisition must be performed, and the analysis of the printed traces must be carried out, ideally avoiding the edges because of possible artifacts. A set of geometrical features is then measured: the height and width of the printed trace of the single layer, the external curvature of the 90° angles, the minimum detectable spacing between the filaments, the overall serpentine height and the deflection of the spanning element. The measured values of each geometric characteristic must be normalized to the values set by the design. Table 2 reports the parameters to be measured together with


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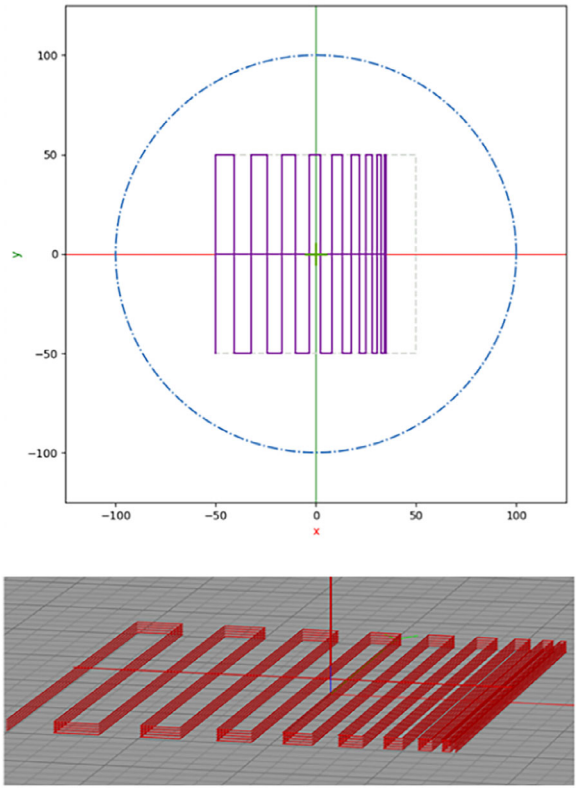


**Figure 9.** Schematical representation of the serpentine test.

the normalized value and the corresponding subfunction. Since these parameters are always measured according to a standardized pattern, they can also be used as metrics to numerically compare the performance of different inks. Moreover, normalization enables the evaluation of printability at different scales, making the methodology suitable for different materials and applications. For this purpose, a software to create custom G-code files, "G-code generator", has been developed and made available on a GitHub repository at the following link: https://github.com/pcomepaolo/gcode_generator. With this application it is possible to define the scale of the serpentine to be printed depending on the features of the printer, such as its printing plate geometry, nozzle diameter and bed size, and desired number of layers to be printed. This test can also be used for the initial calibration of the printer, as described in the documentation of the G-code generator. A careful calibration of the instrument is a necessary prerequisite to gain reliable results from the image analysis of the ink. To calibrate the printer, a single-layer serpentine should be printed with a standard ink (intended as a material in which viscoelastic effect can be neglected, or commercial clays specifically conceived to be modelled by DIW), and the width and height of the deposited layer should be measured to account for over- or under-extrusion. Equations (17), (18) resume the definition of over- and under- extrusion, based on the comparison of the height and width of the single layer ($H_{pr}$ and $W_{pr}$ respectively) with the theoretical layer height and width of the serpentine (H and W respectively). In principle, the printer will over-extrude if both dimensions are higher than those specified by the G-code, and vice versa. The correct extrusion conditions can only be achieved by tuning the printer parameters, which must be saved in the printer profile for further use.

After calibrating the printer, the serpentine test must be performed with the specific ink to be printed, and the ink-to-nozzle speed ratio should be fine-tuned to account for over- and under-extrusion. The evaluation of the extrusion conditions can be obtained by checking the parameters indicated in Equations (17), (18).

$$\text{Over} - \text{extrusion} \quad \text{if} \begin{cases} \mathbf{W}_{pr} > W \\ \mathbf{H}_{pr} > H \end{cases} \tag{17}$$

$$\text{Under} - \text{extrusion} \quad \text{if} \begin{cases} \mathbf{W}_{pr} < W \\ \mathbf{H}_{pr} < H \end{cases} \tag{18}$$

Then, the Single Filament Accuracy is measured using the $|\Delta H|/H$ and $|\Delta W|/W$ ratios (ideally one of the two ratios would be sufficient but measuring both ratios is suggested for better reliability). The lower the ratio, the higher the accuracy of the filament. From the analysis of the corresponding dimensionless number Equation (10), it is easy to relate the results of the printing test to the material properties. Regardless of the driving force that causes a loss of SFA (surface tension or gravity), it is evident that an increase in yield stress can be beneficial to the self-standing ability of the material. Using a suitable additive or tuning the binder composition is sometimes sufficient to achieve the goal.[35] The same approach applies to the Buildability subfunction, measured as the height of the multilayer serpentine and normalized to the theoretical height value. As for the Planar Accuracy, the corresponding dimensionless number Equation (11) suggests acting on elastic modulus, density and surface tension. However, improving PA by tuning the ink formulation may conflict with the optimization of other subfunctions: for example, gravity plays an opposite role when it comes to increasing PA and SFA. To avoid these problems, it may be possible in this case to adjust the printing speed, an additional parameter that helps in optimizing the planar shape without necessarily reformulate the ink composition.[69]

Finally, depositing the suspended filament and measuring the deflections between each span of the serpentine will help to optimize the Suspended Structures subfunction. The dimensionless geometrical feature to be measured is the ratio between the vertical deflection of the truss and the corresponding span. This ratio should be evaluated for each suspended part of the segment. For larger spans, the segment may collapse and break. In this case, the ratio determined for the last span before collapse should be taken as the measure of the subfunction. To enhance it, as described in Equation (16), one can only work on material-related properties: by decreasing the density of the ink or enhancing elastic modulus. On the other hand, one can also work on the design of the object to be printed, trying to reduce the span of the overhangs. It is worth noting that for three out of the subfunctions (i.e., for SFA, B, SS), the material-related parameters such as $\rho$, $\tau_y$, and $G'$, act in the same way, which makes it easier to work with the ink formulation. In contrast, they have the opposite effect on the PA subfunction, which can be improved more efficiently by adjusting process-related parameters such as $v_n$ or the standoff distance.


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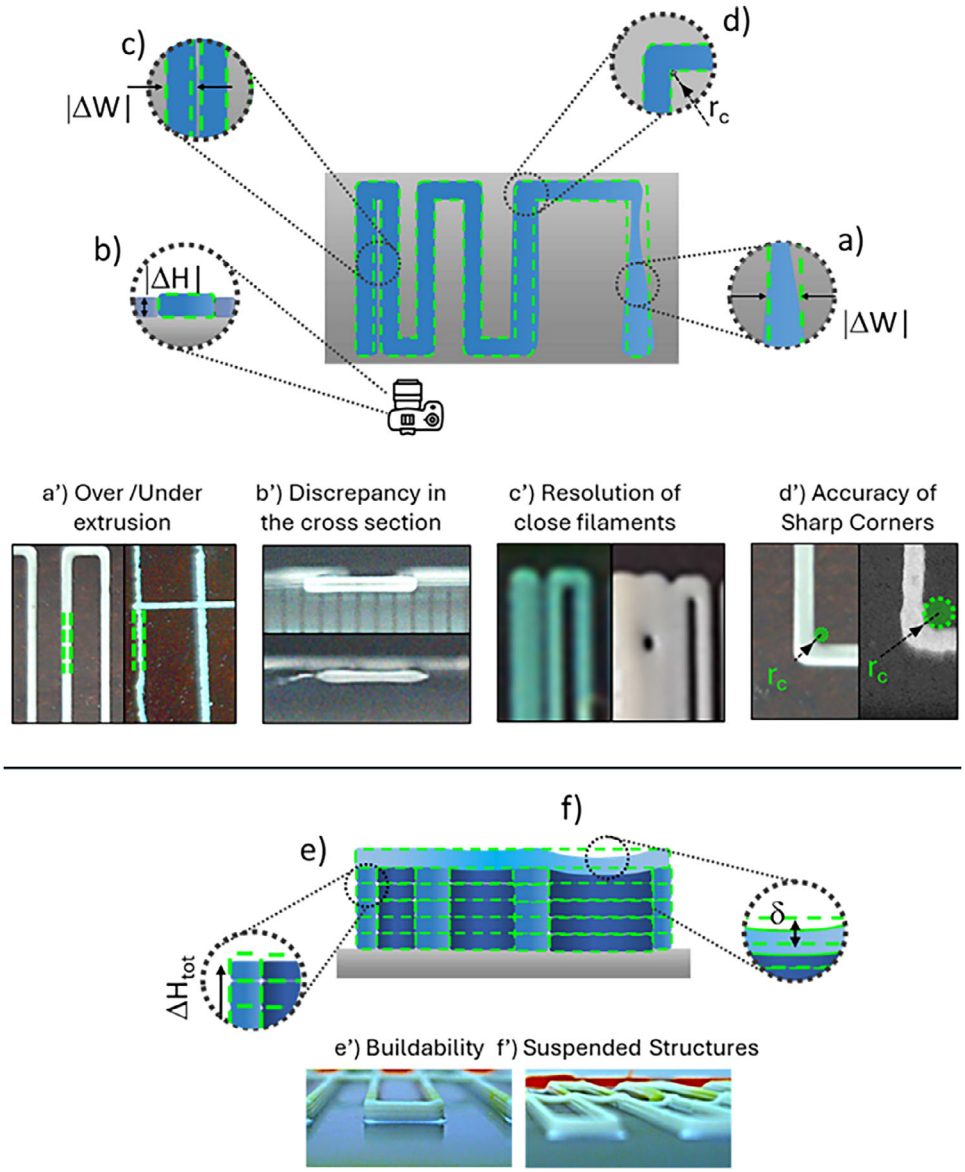


**Figure 10.** a–f) Schematic representation of the serpentine printing test highlighting the defects described by each subfunction: a) over/under extrusion, b) discrepancy with respect to the cross-section imposed by design, c) incapability to print very close yet distinct filaments, d) incapability to reproduce sharp corners, e) collapse of superimposed layers and f) bending of the suspended structures. a–f) Photos depict the same features for a 3D printed serpentine: the left/ upper side and right/bottom side of each picture report a passed and a failed test respectively.

**Figure 10** summarizes, for each of the subfunctions, a passed and a failed test. More in general, the image analysis of the serpentine allows the rating of the inks through a unique and numerical metric, thus overcoming the inconsistencies given by a personal and visual inspection of the materials.

## 5. Conclusion and Final Remarks

Direct Ink Writing (DIW) has emerged as a highly versatile additive manufacturing technique, applicable to a broad range of material systems. The simplicity of DIW printer hardware has undoubtedly facilitated the spread of the technology; however, this same simplicity has contributed to a fragmented and unsystematic accumulation of knowledge regarding its operation and optimization. The critical review of the current literature reveals two main issues: in the first place, the existing printability criteria —typically based on thresholds of rheological properties, such as yield stress or elastic modulus—are often system-specific and fail to generalize across different formulations. In addition, no universally accepted quantitative methodology exists to assess the quality of printed objects, with many studies relying instead on subjective visual evaluations. To address these limitations, we proposed a systematic framework that tries to solve both problems. First, it decomposes the DIW process into a set of fundamental subfunctions: extrudability, single filament accuracy, planar accuracy, buildability, and suspended structure


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capability. Each subfunction was analyzed in detail, discussing the complex relationships between the printed ink and process parameters, and suggesting protocols to rigorously measure the relevant material properties. When explicit correlations could not be derived, dimensional analysis was used to identify the dimensionless groups that govern the physics of each printing step, providing a comprehensive understanding of how specific parameters affect printing quality. Notably, the analysis reveals the complex interdependencies among parameters—some of which may enhance one subfunction while detracting from another—thereby highlighting the need for multi-objective optimization strategies. To be effective, such strategies must also make use of a quantitative protocol to assess the ink printability, thus solving the second issue that arises from the visual inspection of the printed shapes. To do this, we developed a simple yet reliable testing protocol consisting of printing first a single-layer serpentine and then a stacked sequence of serpentines with a suspended filament on top. These shapes are quick to produce but effective in providing information about each subfunction. We also provided normalized metrics to evaluate the printability of the ink in each subfunction by measuring specific geometrical features of the printed shapes. The proposed framework enables objective and reproducible comparisons between different formulations, allows for systematic identification and correction of printing quality problems, and will hopefully establish a common standard for the rational design and optimization of better inks.

In conclusion, DIW printability cannot be universally characterized by fixed rheological thresholds. Instead, it should be understood as the synergistic result of multiple interrelated subfunctions, each influenced by specific physical variables. A unified, dimensionless framework – combined with quantitative, reproducible testing – offers a robust foundation for optimizing materials and processes in DIW.

## Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

## Conflict of Interest

The authors declare no conflict of interest.

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3D printing, additive manufacturing, ink optimization, printability test, rheology

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